

RESEARCH EXPERIENCE

- **Multi-Modal Foundational Model** 2023 - Present
Video/Image Diffusion Model; Multi-Modal Large Language Model
- **Gaussian Splatting** 2023 - Present
World Model; Surgical Simulation; Watermarking
- **Object Detection in the Open World** 2020 - 2024
Domain Adaptation; Open-Vocabulary/Set/World Object Detection; Noisy Labels
- **Graph Learning** 2020 - 2024
Graph Matching; Causal Graph; Graph Motif
- **Medical Imaging Analysis** 2020 - 2024
MRI Brain Analysis; Tooth Analysis; Medical Report Generation

WORK EXPERIENCE

- **The Chinese University of Hong Kong** Feb. 2024 - Jan. 2025
Postdoc, Electronic Engineering
Supervisor: Prof. Yixuan Yuan

EDUCATION

- **City University of Hong Kong** Aug. 2020 - Dec. 2023
Ph.D., Electrical Engineering
Supervisor: Prof. Yixuan Yuan
Early Graduation
- **Tianjin University** Sep. 2016 - Jun. 2020
B.S., Electrical Engineering

SELECTED CONFERENCE PAPER (‘[]’ HIGHLIGHTS KEY WORDS)

1. **[Diffusion] CLIFF: Continual Latent Diffusion for Open-Vocabulary Object Detection**
Wuyang Li, Xinyu Liu, Jiayi Ma, Yixuan Yuan
European Conference on Computer Vision (**ECCV**), 2024, **Oral (1.8%)**
2. **[Video Generation] Boosting Medical Image Analysis via Motion-Informed Generative Videos**
Wuyang Li, Xinyu Liu, Qiushi Yang, Yixuan Yuan
Medical Image Computing and Computer Assisted Intervention (**MICCAI**), 2024
3. **[Open-World Learning] Novel Scenes & Classes: Towards Adaptive Open-set Object Detection**
Wuyang Li, Xiaoqing Guo, Yixuan Yuan
IEEE International Conference on Computer Vision (**ICCV**), 2023, **Oral (1.8%)**
4. **[Open-World Learning] Adjustment and Alignment for Unbiased Open Set Domain Adaptation**
Wuyang Li, Jie Liu, Bo Han, Yixuan Yuan
IEEE Conference on Computer Vision and Pattern Recognition (**CVPR**), 2023
5. **[Domain Shift] SIGMA: Semantic-complete Graph Matching for Domain Adaptive Object Detection**
Wuyang Li, Xinyu Liu, Yixuan Yuan
IEEE Conference on Computer Vision and Pattern Recognition (**CVPR**), 2022, **Best Paper Finalist (0.4%)**
6. **[Domain Shift] SCAN: Cross Domain Object Detection with Semantic Conditioned Adaptation**
Wuyang Li, Xinyu Liu, Xiwen, Yao, Yixuan Yuan
The Association for the Advance of Artificial Intelligence (**AAAI**), 2022, **Oral (4.5%)**
7. **[Backbone] U-KAN Makes Strong Backbone for Medical Image Segmentation and Generation**
Chenxin Li*, Xinyu Liu*, **Wuyang Li***, Cheng Wang*, Hengyu Liu, Yixuan Yuan
Under Review, **AAAI**, 2025
8. **[SAM] Flaws can be Applause: Unleashing Potential of Segmenting Ambiguous Objects in SAM**
Chenxin Li*, Yuzhi Huang*, **Wuyang Li**, Hengyu Liu, Xinyu Liu, Qing Xu, Zhen Chen, Yue Huang, Yixuan Yuan
Conference on Neural Information Processing Systems (**NeurIPS**), 2024
9. **[3D] GaussianStego: A Generalizable Stenography Pipeline for Generative 3D Gaussians Splatting**
Chenxin Li, Hengyu Liu, Zhiwen Fan, **Wuyang Li**, Yifan Liu, Panwang Pan, Yixuan Yuan
Under Review, **ICLR**, 2025

10. **[SAM]** When 3D Partial Points Meets SAM: Tooth Point Cloud Segmentation with Sparse Labels
Yifan Liu, Wuyang Li, Cheng Wang, Hui Chen, Yixuan Yuan
Medical Image Computing and Computer Assisted Intervention (**MICCAI**), 2024, **Early Accept (14.0%)**
11. **[Diffusion]** Latent Diffusion Label Rectification for Semi-supervised Medical Image Segmentation
Xinyu Liu, Wuyang Li, Yixuan Yuan
Medical Image Computing and Computer Assisted Intervention (**MICCAI**), 2024
12. **[Brain]** FMRI to T1w MRI Knowledge Transfer Network for Brain Multi-phenotype Prediction
Zhibin He, Wuyang Li, Yu Jiang, Zhihao Peng, Pengyu Wang, Xiang Li, Tianming Liu, Junwei Han, et al.
Medical Image Computing and Computer Assisted Intervention (**MICCAI**), 2024
13. **[3D]** LGS: A Light-weight 4D Gaussian Splatting for Efficient Surgical Scene Reconstruction
Hengyu Liu*, Yifan Liu*, Chenxin Li*, Wuyang Li, Yixuan Yuan
Medical Image Computing and Computer Assisted Intervention (**MICCAI**), 2024
14. **[Video Generation]** Endora: Video Generation Models as Endoscopy Simulators
Chenxin Li, Hengyu Liu, Yifan Liu, Brandom Feng, Wuyang Li, Xinyu Liu, Zhen Chen, Jing Shao, Yixuan Yuan
Medical Image Computing and Computer Assisted Intervention (**MICCAI**), 2024
15. **[Multi-Modality]** MRM: Masked Relation Modeling for Medical Image Pre-Training with Genetics
Qiushi Yang, Wuyang Li, Baopu Li, Yixuan Yuan
IEEE International Conference on Computer Vision (**ICCV**), 2023
16. **[Brain]** H2GM: A Hierarchical Hypergraph Matching Framework for Brain Landmark Alignment
Zhibin He, Wuyang Li, Tuo Zhang, Yixuan Yuan
Medical Image Computing and Computer Assisted Intervention (**MICCAI**), 2023
17. **[Noisy Labels]** Towards Robust Adaptive Object Detection under Noisy Annotations
Xinyu Liu, Wuyang Li, Qiushi Yang, Baopu Li, Yixuan Yuan
IEEE Conference on Computer Vision and Pattern Recognition (**CVPR**), 2022
18. **[Noisy Labels]** Intervention & Interaction Federated Abnormality Detection with Noisy Clients
Xinyu Liu, Wuyang Li, Yixuan Yuan
Medical Image Computing and Computer Assisted Intervention (**MICCAI**), 2022, **Oral**

SELECTED JOURNAL PAPER (‘[]’ HIGHLIGHTS KEY WORDS)

1. **[Multi-Modality]** Bridge the Gap Between Visual and Linguistic Comprehension for Zero-shot Semantic Segmentation
Xiaoqing Guo*, Wuyang Li*, Yixuan Yuan
Under Revision, Information Fusion, 2024
2. **[Domain Shift]** SIGMA++: Improved Semantic-complete Graph Matching for Domain Adaptive Object Detection
Wuyang Li, Xinyu Liu, Yixuan Yuan
IEEE Transactions on Pattern Analysis and Machine Intelligence (**TPAMI**), 2023
3. **[Object Detection]** HTD: Heterogeneous Task Decoupling for Two-Stage Object Detection
Wuyang Li, Zhen Chen, Baopu Li, Dingwen Zhang, Yixuan Yuan
IEEE Transactions on Image Processing (**TIP**), 2021
4. **[3D]** Stereo Vision Meta-Lens-Assisted Driving Vision
Xiaoyuan Liu, Wuyang Li, Takeshi Yamaguchi, Zihan Geng, Takuo Tanaka, Din Ping Tsai, Mu Ku Chen
ACS Photonics, 2024, **Selected as Cover Paper**
5. **[Backbone]** Harnessing Lightweight Transformer with Contextual Synergic Enhancement for Efficient 3D Medical Image Segmentation
Xinyu Liu, Zhen Chen, Wuyang Li, Yixuan Yuan
Under Review, TPAMI, 2024
6. **[LLM]** LLM-guided Decoupled Probabilistic Prompt for Continual Learning in Medical Diagnosis
Yiwen Luo, Wuyang Li, Chen Cheng, Xiang Li, Tianming Liu, Yixuan Yuan
Under Revision, TMI, 2024
7. **[Brain]** Foundation Model for Any Phenotype Prediction via fMRI to sMRI Knowledge Transfer
Zhibin He, Wuyang Li, Yifan Liu, Xinyu Liu, Junwei Han, Tuo Zhang, Yixuan Yuan
Under Revision, TMI, 2024

8. **[Noisy Labels]** Medical Federated Learning with Joint Graph Purification for Noisy Label Learning
Zhen Chen, **Wuyang Li**, Xiaohan Xing, Yixuan Yuan
Medical Image Analysis (**MIA**), 2023
9. **[Causality]** Decoupled Unbiased Teacher for Source-free Domain Adaptive Medical Object Detection
Xinyu Liu, **Wuyang Li**, Yixuan Yuan
IEEE Transactions on Neural Networks and Learning Systems (**TNNLS**), 2023
10. **[3D]** GRAB-Net: Graph-based Boundary-aware Network for Medical Point Cloud Segmentation
Yifan Liu, **Wuyang Li**, Jie Liu, Hui Chen, Yixuan Yuan
IEEE Transactions on Medical Imaging (**TMI**), 2023

SELECTED AWARDS

• Outstanding Academic Performance Award, CityU Outstanding research achievements	2022 - 2023
• Research Tuition Scholarship, CityU Outstanding research achievements	2022 - 2023
• Ranked First in Polyp Detection Challenge 3rd International Endoscopy Computer Vision Challenge and Workshop (EndoCV2021)	2021
• National Scholarship The great honor for a student in China	2017 - 2018
• Silver Award in Tianjin Advanced Mathematical Competition As a selected participant representing Tianjin University	2017

SERVICES

• NeruIPS-24 Workshop Organizer [Link] Assisted the <i>Advancements in Multi-modal Medical Foundation Models</i> workshop [Proposal Link]	2024
• Research Funding Proposal Assisted my advisor in completing five research funding proposals over the past two years	2022 - 2024
• Startup Funding Proposal [Link] Successfully secured seed funding from HK Tech 300 and HKSTP	2023
• Founder of ScholaGO Education Technology Company Limited, Hong Kong [Link] Founded a company to develop a multi-modal educational product	2024
• Conference and Journal Reviewer T-PAMI; IJCV; TIP; TNNLS; NeurIPS-24; CVPR-23/24; ICCV-23; AAAI-23/24; NeurIPS-25; ICLR-25	